Cryptocurrency Fundamentals

Module 2 Study Guide

How Cryptocurrencies Work

This guide covers every lesson of the module, using only words and figures from the lessons themselves. Work through it after watching the lessons, then use the checklist at the end to test yourself.

Lesson 1: Coins, Tokens, and Digital Ownership

In this lesson: Native Coins vs Tokens; Smart Contracts & Fees; TT$ Wallet Worked Example; P

Key points

- Coins operate on native blockchains while tokens use host networks
- Moving tokens requires holding native coins for network fees
- Private keys grant true cryptographic ownership of your digital assets

Examples from the lesson

- In our course, Cryptocurrency Fundamentals, we are kicking off Module 2, "How Cryptocurrencies Work," with Lesson 1: "Coins, Tokens, and Digital Ownership."
- ERC-20 tokens on Ethereum or tokens on Solana do not maintain their own independent blockchains; instead, they rely entirely on the host network to process and record their transactions.
- Dev decides to buy TT$ 1,000 worth of Ethereum, which is a native coin, to fund his digital wallet.
- Next, he purchases TT$ 500 worth of a utility token built directly on the Ethereum network.
- His wallet now holds two distinct types of digital assets that function differently under the hood.
- To initiate a transfer of TT$ 200 worth of his utility tokens to a friend in Port of Spain, Dev cannot pay the network fee using those tokens.
- Dev pays a network fee equivalent to TT$ 15 using his native coin balance to complete the transaction.
- Dev starts with TT$ 1,000 in native coins and buys TT$ 500 in utility tokens, bringing his initial total portfolio value to TT$ 1,500.
- After sending TT$ 200 worth of tokens and paying the TT$ 15 network fee in native coins, his remaining token balance is TT$ 300 and his remaining native coin balance is TT$ 985.
- His total remaining wallet balance stands at TT$ 1,285.
- In our next lesson, Lesson 2: "How Transactions Move From Sender to Receiver," we will look under the hood to see how a transaction travels from your phone to the blockchain ledger.

Lesson 2: How Transactions Move From Sender to Receiver

In this lesson: Step 1 & 2: Initiation & Signing; Step 3 & 4: Broadcasting & Mempool; Step 5 &

Key points

- Wallet initiation and key signing start the process
- Broadcasting nodes and mempools manage waiting transactions
- Block validation permanently records confirmed balance updates

Examples from the lesson

- In our course Cryptocurrency Fundamentals, Module 2: How Cryptocurrencies Work, today we are jumping into Lesson 2: How Transactions Move From Sender to Receiver.
- Step one begins with transaction initiation in your digital wallet.
- Let us say you owe a friend TT$100 for your share of maxi taxi fare and lunch during a liming session.
- In your wallet app, you type in your friend’s wallet address, enter the amount of TT$100, and click send.
- Step two is digital signing using your private key.
- Your wallet app automatically creates a cryptographic signature for this TT$100 transfer, ensuring no one can forge your instruction or change the recipient.
- Step three is broadcasting the transaction to the peer-to-peer network.
- In a flash, nodes across Trinidad and Tobago and around the world receive the message and double-check that you have enough funds to cover the TT$100.
- Step four is entering the unconfirmed transaction pool, often called the mempool.
- Your TT$100 transfer waits in line alongside thousands of other global transactions until network validators pick it up.
- Step five is block assembly and validation by network miners or validators.
- These computers bundle your TT$100 transaction into a cryptographic block of data, alongside other transactions waiting in the mempool.
- Step six is final block confirmation and ledger update.
- Your friend receives a notification that the TT$100 transaction is fully confirmed, and the ledger permanently reflects the new balance.
- You start with a digital wallet balance of TT$500 in your account before the transfer.
- You initiate a transaction send of TT$100 to pay back your lime partner.
- The network charges a tiny processing fee of TT$2 for validation.
- This leaves your total deducted amount at TT$102 from your original wallet.
- Meanwhile, your friend starts with a balance of TT$200 in their digital wallet.
- Your friend receives the exact sent amount of TT$100 credited directly into their account.
- After network confirmation, your ending wallet balance is TT$398 and your friend’s ending wallet balance is TT$300.
- A crypto transaction moves through six distinct steps: initiation in your wallet, signing with your private key, broadcasting to nodes, waiting in the mempool, block assembly by

validators, and final block confirmation on the blockchain ledger.

- In our next lesson, Lesson 3: Public Keys, Private Keys, and Addresses, we will look closer at the cryptographic keys that keep your account secure.

Lesson 3: Public Keys, Private Keys, and Addresses

In this lesson: Mailbox vs Master Key; Public Keys & Addresses; Private Key Security; Worked T

Key points

- Public addresses are shared freely to receive funds
- Private keys authorize transfers and must stay completely secret
- Darren transferred TT$340 safely by signing without exposing his private key

Examples from the lesson

- I’m glad you’re here for Cryptocurrency Fundamentals, Module 2: How Cryptocurrencies Work.
- Today we are diving into Lesson 3: Public Keys, Private Keys, and Addresses.
- Darren wants to send TT$340 worth of cryptocurrency to his supplier, Suresh, to buy new seedling trays and organic fertilizer for his farm.
- Darren enters Suresh’s address into his own wallet and enters the payment amount of TT$340.
- His private key signs the outgoing transfer of TT$340 behind the scenes without ever revealing the key itself to Suresh or the public network.
- Darren’s wallet balance decreases by TT$340, while Suresh’s address shows a new incoming credit of TT$340.
- Darren’s TT$340 transaction succeeded safely because his private key signed the transfer without ever exposing his master credentials to the network.

Recap checklist — can you say each one?

Lesson 1: Coins, Tokens, and Digital Ownership

- ☐ Coins operate on native blockchains while tokens use host networks
- ☐ Moving tokens requires holding native coins for network fees
- ☐ Private keys grant true cryptographic ownership of your digital assets

Lesson 2: How Transactions Move From Sender to Receiver

- ☐ Wallet initiation and key signing start the process
- ☐ Broadcasting nodes and mempools manage waiting transactions
- ☐ Block validation permanently records confirmed balance updates

Lesson 3: Public Keys, Private Keys, and Addresses

- ☐ Public addresses are shared freely to receive funds
- ☐ Private keys authorize transfers and must stay completely secret
- ☐ Darren transferred TT$340 safely by signing without exposing his private key